# AI-Based Diabetes Risk Prediction Research Report

***Machine Learning & Lifestyle Factors Across Multi-Cohort Healthcare Data***

## 1. Executive Summary & Study Overview

This research investigates how modifiable lifestyle habits, demographic characteristics, and laboratory biomarkers predict diabetes status across six benchmark datasets comprising over 350,000 records. Using scikit-learn, LightGBM, XGBoost, and SHAP, we engineered specialized, leakage-free pipelines across clinical, survey, and acute symptom cohorts.

## 2. Data Integrity & Cryptographic Audit

| Cohort / Dataset | Records | Features | Duplicates | Target |
|---|---|---|---|---|
| Clinical EHR (Primary) | 100,000 | 9 | 3,854 (3.8%) | diabetes (8.5% pos) |
| Clinical EHR (Copy) | 100,000 | 9 | Identical MD5 | Duplicate - Excluded |
| CDC BRFSS Survey (Binary) | 253,680 | 22 | 24,206 (9.5%) | Diabetes_binary (13.9% pos) |
| CDC BRFSS (Balanced 5050) | 70,692 | 22 | 1,635 (2.3%) | Diabetes_binary (50.0% pos) |
| CDC BRFSS (3-Class 012) | 253,680 | 22 | 23,899 (9.4%) | Diabetes_012 (0/1/2) |
| Sylhet Early Symptoms | 520 | 17 | 269 (51.7%) | class (61.5% pos) |

## 3. Model Benchmark & Evaluation Summary

| Cohort | Selected Model | Accuracy | Sensitivity | ROC-AUC | Calibrated Brier |
|---|---|---|---|---|---|
| Clinical Cohort | LightGBM (Calibrated) | 94.2% | 92.1% | 0.9770 | 0.0236 |
| BRFSS Lifestyle | LightGBM (Balanced) | 73.8% | 74.5% | 0.8046 | 0.1080 |
| Early Symptoms | Extra Trees (Calibrated) | 99.0% | 100.0% | 0.9996 | 0.0074 |
| BRFSS Balanced | XGBoost (Calibrated) | 73.9% | 80.2% | 0.8192 | 0.1726 |

## 4. Scientific Ablation: Clinical Biomarkers vs. Lifestyle

Ablation testing demonstrated that while lifestyle factors alone provide a valuable non-invasive screening baseline (ROC-AUC 0.825), the addition of laboratory glycemic markers (HbA1c & Blood Glucose) provides a massive +0.152 increase in ROC-AUC (reaching 0.977) and lowers the false-negative rate from 20.3% to 7.9%.

## 5. Explainable AI & Feature Attribution

Using SHAP TreeExplainer, HbA1c and Blood Glucose were isolated as dominant laboratory predictors. For survey-based lifestyle prediction, self-rated general health (GenHlth), hypertension, BMI, age, and physical activity formed the primary risk signal. In acute triage, polyuria and polydipsia dominated model attribution.

## 6. Critical Limitations & Governance

IMPORTANT DISTINCTION: CURRENT STATUS VS. FUTURE INCIDENCE. The supplied datasets contain retrospective cross-sectional health surveys and clinical records labeled with existing diabetes status. They do NOT contain prospective longitudinal tracking. Consequently, these models predict current classification likelihood consistent with survey or clinical diagnostic criteria, rather than guaranteeing future disease onset.

**ETHICAL NOTICE:** RESEARCH & EDUCATIONAL USE ONLY. This system is an artificial intelligence research prototype designed to explore machine learning associations with diabetes status. It is NOT a certified medical diagnostic device and does NOT provide medical advice, diagnosis, treatment, or prevention of any medical condition. The predicted probabilities reflect classification patterns observed in retrospective public datasets, not clinically calibrated absolute lifetime risks. Always consult a qualified physician or healthcare provider for any medical questions, clinical evaluation, or laboratory testing (e.g., HbA1c, oral glucose tolerance test).